

Aryan Kumar

✉ kraryan700@gmail.com

☎ +91-8987041676

📍 Bangalore, Karnataka

🌐 linkedin.com/in/aryan700

🐙 github.com/aryankr2005

🔗 SKILLS

3D modelling and Simulation

- SolidWorks
- Fusion360
- AutoCad
- Ansys Workbench

📄 CERTIFICATIONS

- Certified Solidworks Associate (CSWA)
- AutoCad 2D & 3D (Udemy)
- Generative Design in Fusion360

🎓 EDUCATION

B.Tech Mechanical Engineering

NIT Tiruchirappalli
2023 – present

Senior Secondary Education

St. Dominic Savio's High School
2021 – 2023

🏆 ACHIEVEMENTS

- Bajaj TORQ '25 National Finalists

📁 RESEARCH INTERNSHIP

Evaluating Whole Body Vibration Effects on the Lumbar Spine using FEM

IIT Hyderabad

05/2025

- Developed a detailed finite element model of the **human lumbar spine** based on MRI data and **50th percentile male** including vertebrae and intervertebral discs.
- Conducted **static** and **dynamic** simulations to analyse **biomechanical behaviour**, performing range of motion, intradiscal pressure, and vibration analyses to validate the model and study lumbar spine responses under various loading and whole-body vibration conditions.
- Provided insights into lumbar spine injury mechanisms under vibration by identifying stress and strain patterns in the lumbar discs, highlighting **regions prone to damage** and offering information valuable for improving **driver health and ergonomic design**.

📁 PROJECTS

AUTONOMOUS MOBILE ROBOT (AMR)

Designed and developed a compact AMR

06/2026

- Designed a compact **AMR** in Solidworks meeting constraints of **500×500×500mm, 20kg payload**.
- Developed a **2-drive, 4-caster wheel architecture** to achieve high maneuverability and stable load distribution.
- Optimized chassis and internal packaging for integration of battery, electronics, sensors, and safety components, achieving a total operating mass of **38.47 kg**.
- Applied **load, clearance, and structural calculations** with **DFMA and material selection** to deliver a lightweight, manufacturable design.

SILICO DAMP

Low Noise Motorcycle Stand Dampener

03/2025

- Designed and simulated a **silicone-based dampening solution** using **CAD/CAE** tools to reduce noise from motorcycle stand impacts by increasing strain energy absorption.
- Conducted cost-benefit and manufacturability analysis comparing standard and **high-damping silicone rubber pads**; proposed **compression molding** for scalable production.
- Recommended process optimizations including injection molding, mold redesign to **reduce manufacturing costs by ~20%** while maintaining product performance.

MOTOPULSE

Electric Motor Conversion for Normal cycles

02/2025

- Designed a **24V, 250W BLDC mid drive motor-based** system in **Solidworks** to convert regular bicycles into electric-powered ones, making them more efficient and accessible.
- Engineered a **Custom Spindle and Shaft Lock Mechanism** by extending the main spindle and adding two hollow shafts for the pedals, allowing the right pedal to remain independent of the main crank. A **ball bearing system** inside the hollow shaft enabled smooth movement, while a **push-pin lock mechanism**.
- Designed a system where riders could **switch between electric and manual pedalling** by simply locking or unlocking the spindle, ensuring smooth power delivery, better efficiency, and a natural riding experience.